

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 8

STRUCTURE OF ATMOSPHERE

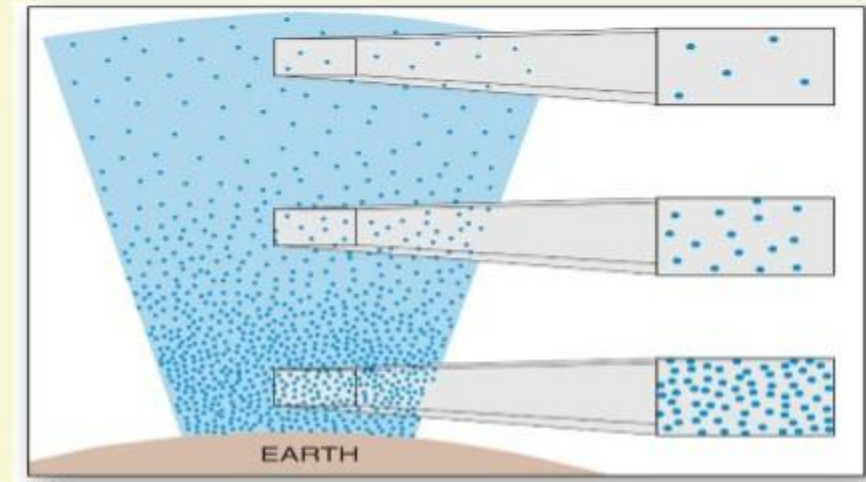
- **EXOSPHERE**
- **CHEMICAL COMPOSITION OF ATMOSPHERE**

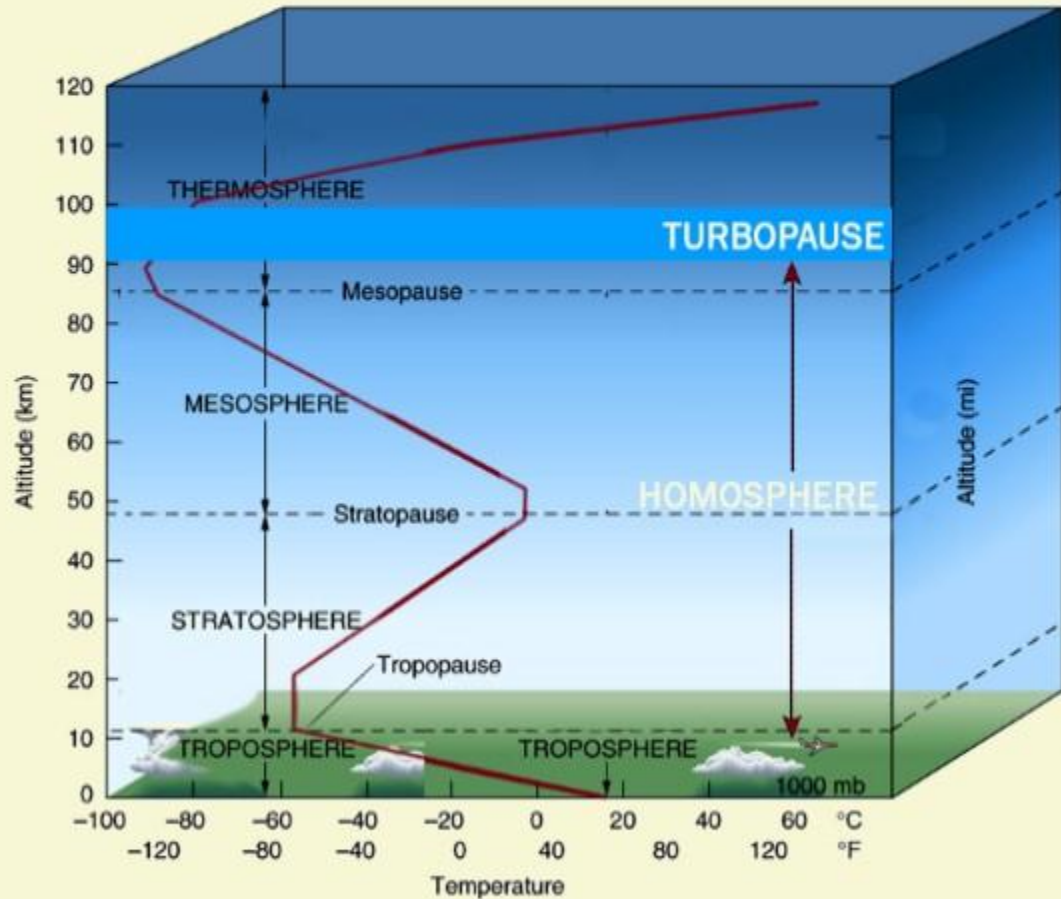
STRUCTURE OF THE ATMOSPHERE

5 EXOSPHERE



- 1 This is the uppermost layer of the earth's atmosphere.
- 2 Here the atmosphere merges into space.
- 3 Highly rarified atmosphere





HOMOSPHERE

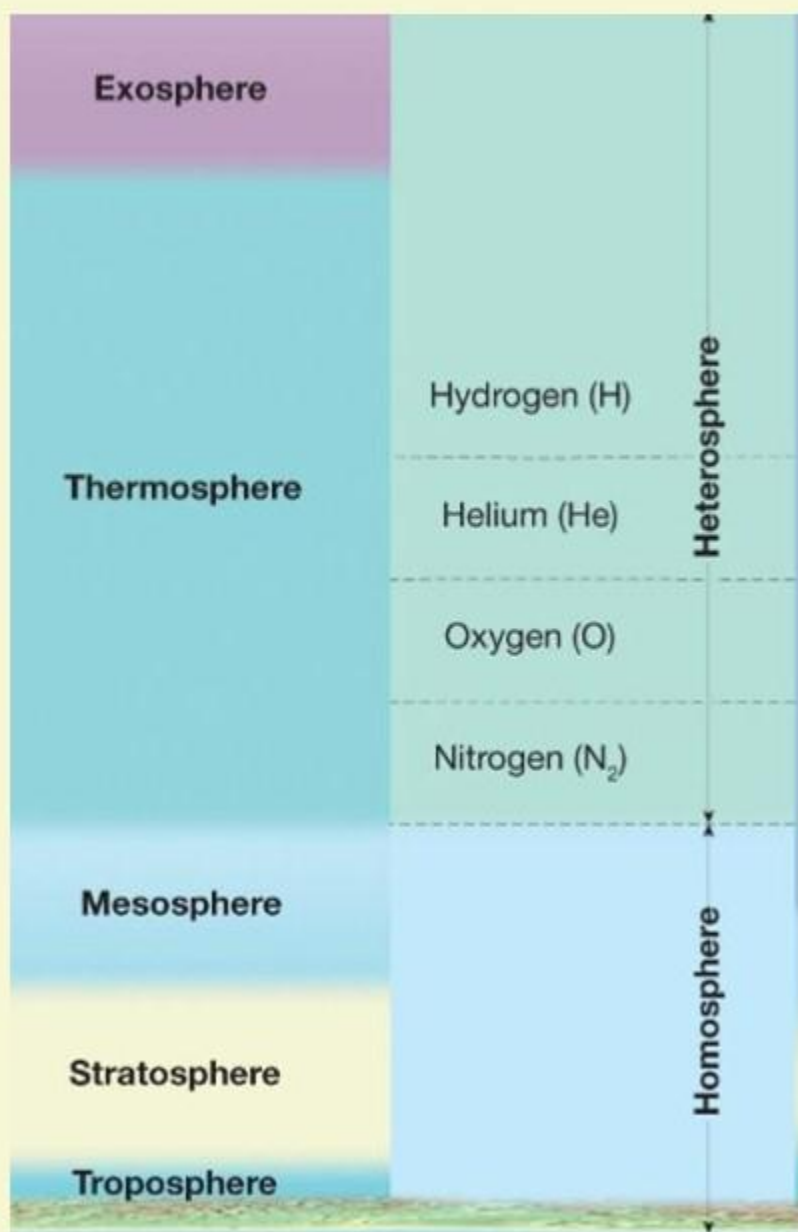
-Turbulent
Mixing

HOMOSPHERE

- 1 The homosphere is a zone of the atmosphere where the constituent gases are **fairly well mixed**.
- 2 Homosphere consists of layers till 80km-troposphere, stratosphere and mesosphere layers.

WHAT IS TURBOPAUSE?

The turbopause marks the altitude in the Earth's atmosphere below which **turbulent mixing dominates**.



HETEROSPHERE

-Stratification as per molecular weight

HETEROSPHERE

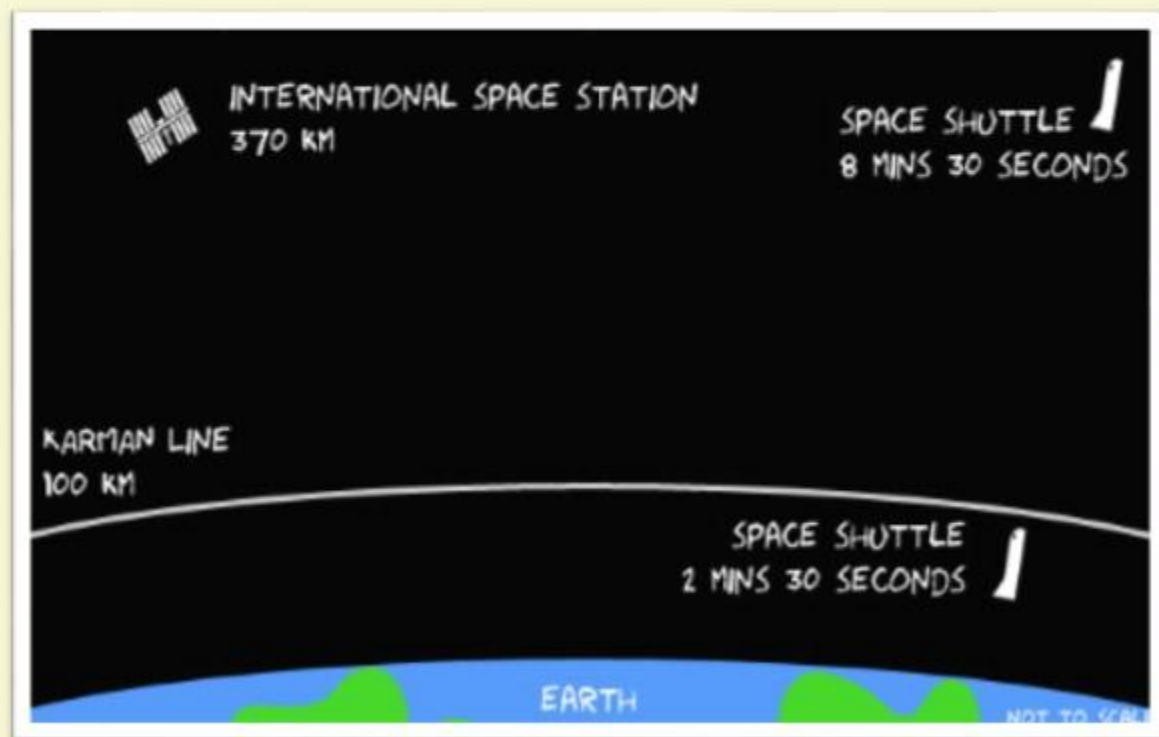
1 The heterosphere lies above the homosphere and is a **zone of poor mixing** where the average composition varies depending on the location within it.

2 Heterosphere consists of:

- Molecular Nitrogen
- Atomic Oxygen Layer
- Helium Layer
- Atomic Hydrogen Layer

FEW MORE IMPORTANT CONCEPTS

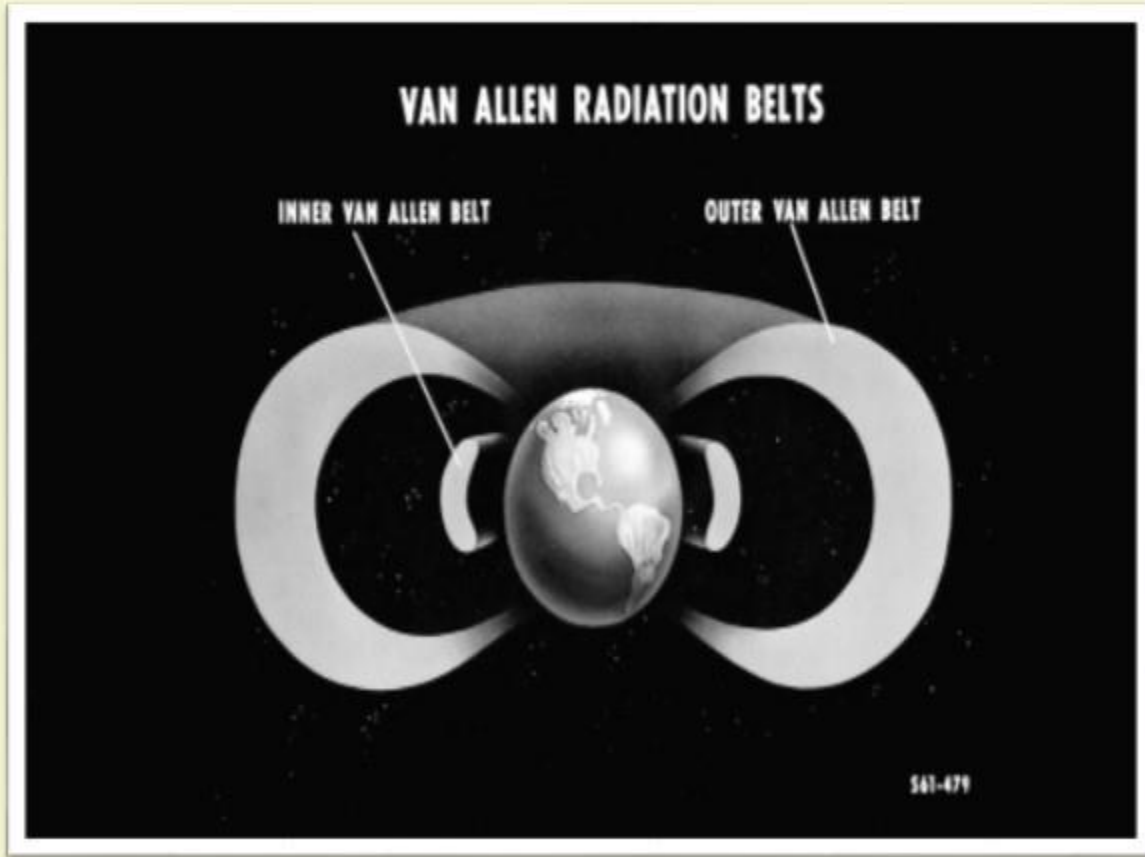
KARMAN LINE



- 1 This line represents the boundary between Earth's atmosphere and outer space.
- 2 The line is named after Theodore von Kármán (1881–1963), a Hungarian American engineer and physicist, who was active primarily in aeronautics and astronautics.
- 3 The line is approximately at the turbopause, at an altitude of about 100km, above which atmospheric gases are not well-mixed.

FEW MORE IMPORTANT CONCEPTS

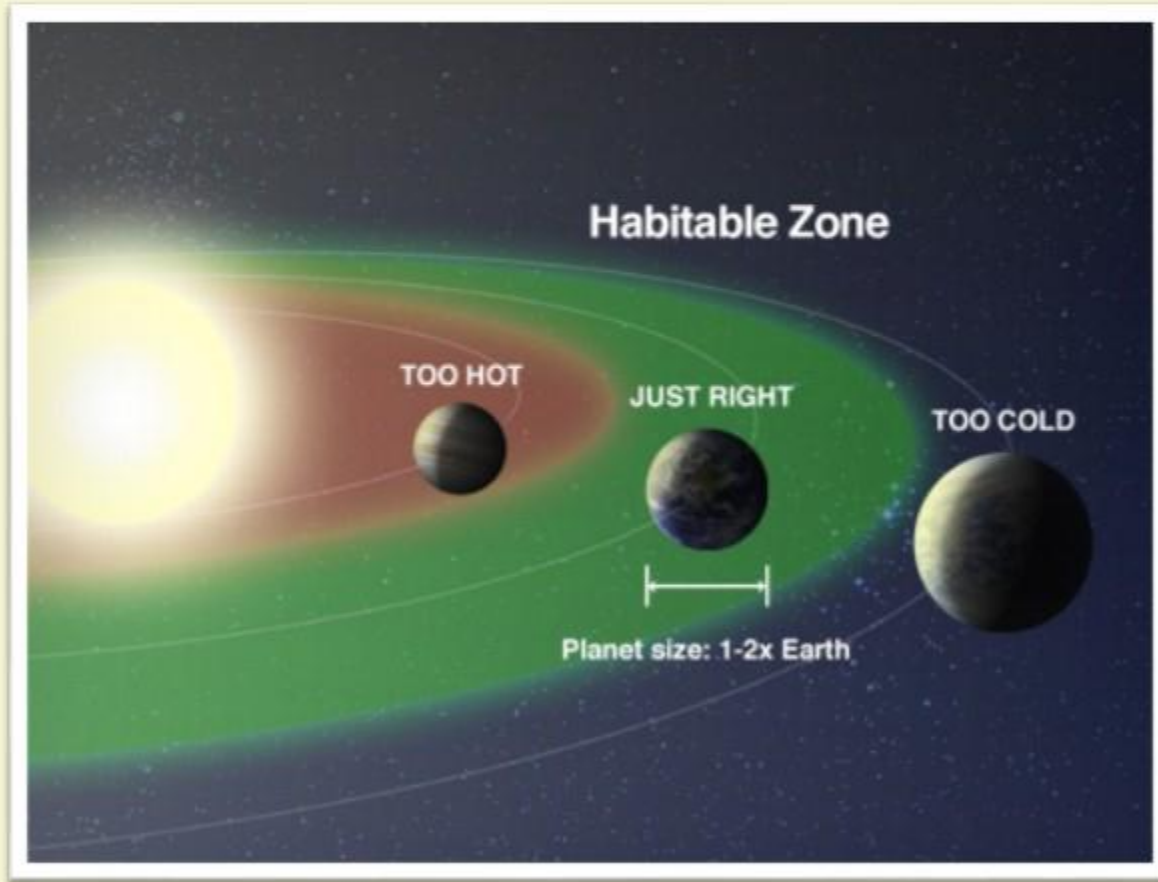
VAN ALLEN RADIATION BELT



- 1** The discovery of the belts is credited to **James Van Allen**.
- 2** Van Allen radiation belt is a **zone of energetic charged particles**, most of which originate from the **solar wind** that is captured by Earth's magnetic field.
- 3** They're held in place by the magnetic field of the Earth, and so they **trace the shape of the magnetic field itself**.

FEW MORE IMPORTANT CONCEPTS

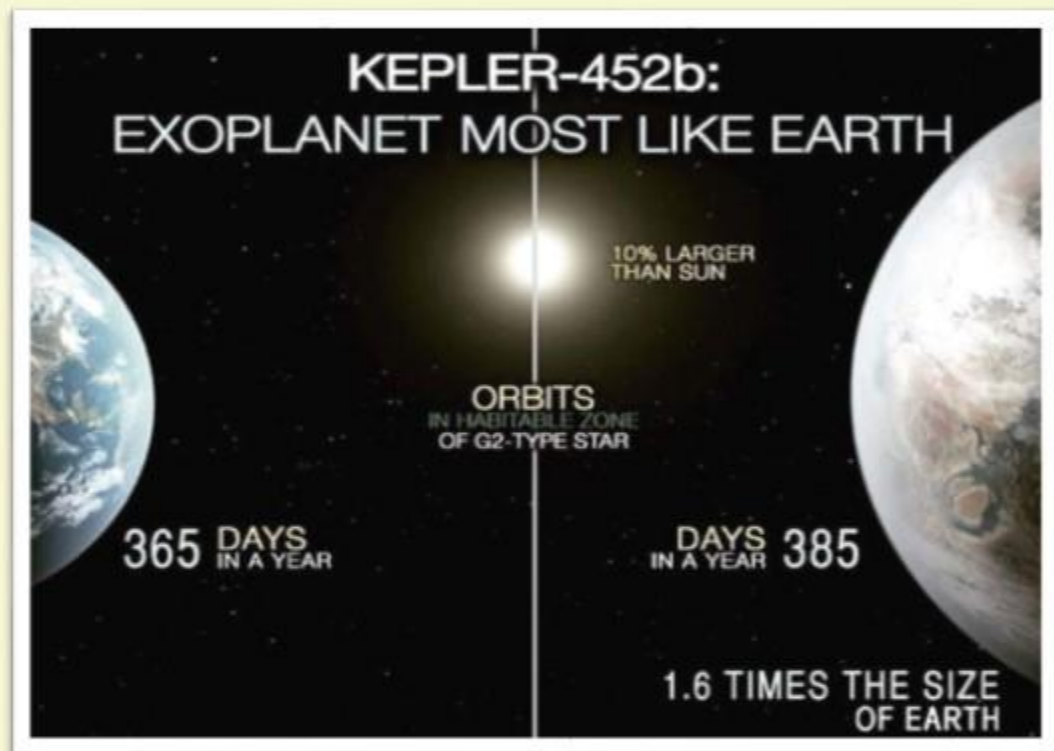
GOLDILOCKS ZONE



- 1** It refers to a habitable zone in the planetary system where the temperature is neither too high nor too low.
- 2** Such conditions could allow for the presence on the planet's surface of liquid water – a key ingredient for life
- 3** If a planet is too close to the star it orbits, any water on the surface quickly boils off, forming a steam atmosphere. If the planet is too far from the star, any water on the surface freezes.

FEW MORE IMPORTANT CONCEPTS

GOLDILOCKS ZONE



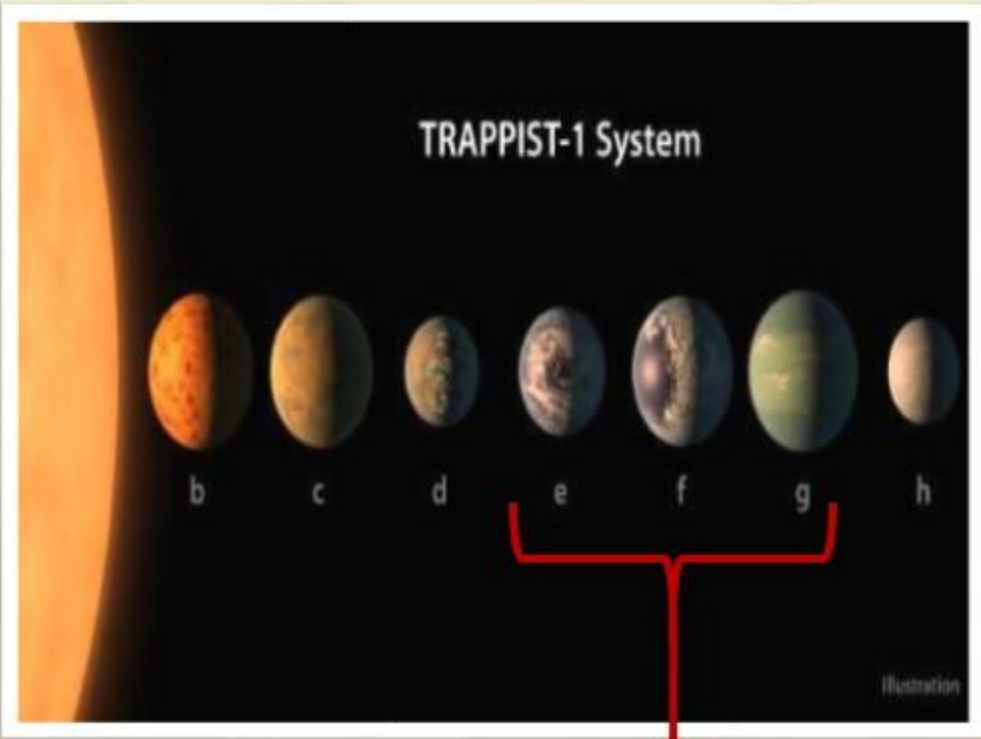
- 1 Earth is in the habitable zone of the sun – one of the reasons our planet has liquid water like oceans and lakes.
- 2 The habitable zone (or “Goldilocks zone”) is the **range of orbital distances from a star at which liquid water can exist on the surface of a planet**
- 3 This range of distances changes **depending on the size and temperature of the star.**

WHAT ARE EXOPLANETS?

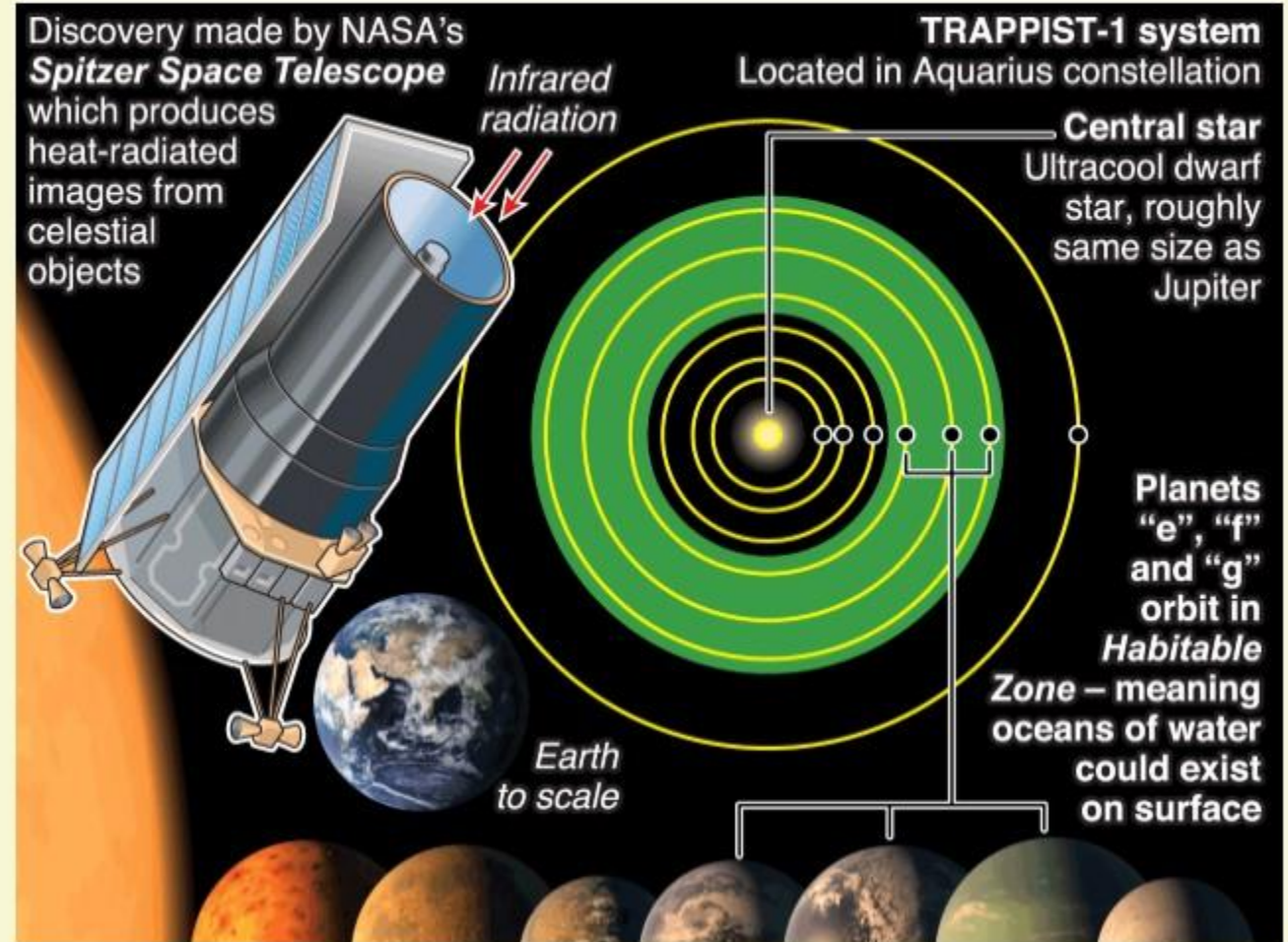
It is a planet that **does not orbit the Sun and instead orbits a different star.** It is also termed as **extrasolar planet.**

FEW MORE IMPORTANT CONCEPTS

GOLDILOCKS ZONE



Planets “e”, “f”, “g” are the most habitable ones.



FEW MORE IMPORTANT CONCEPTS

GOLDILOCKS ZONE

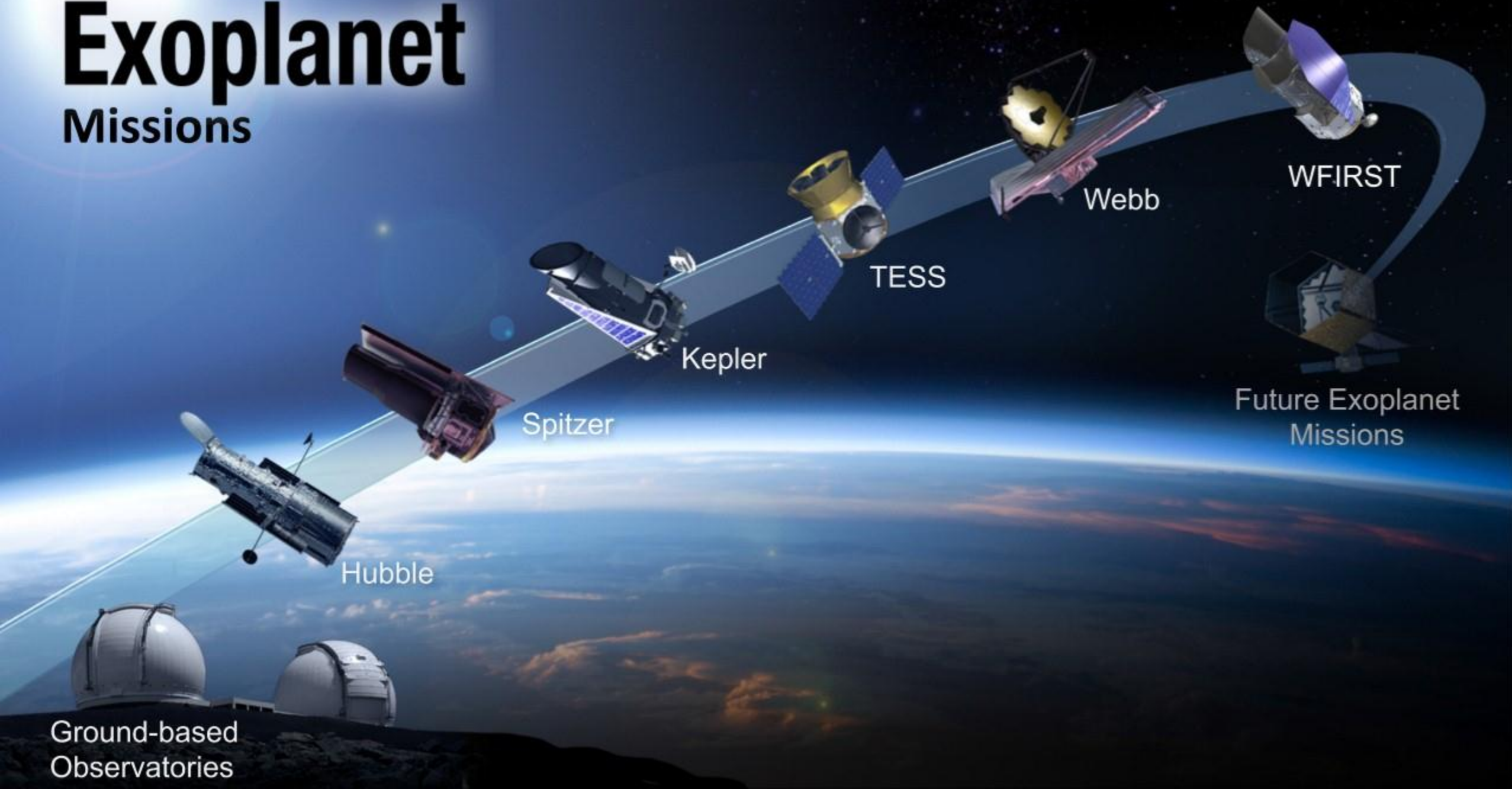
The term 'Goldilocks Zone' is often seen news in the context of (UPSC Prelims 2015)

- (a) The limits of habitable zone above the surface of the Earth
- (b) Regions inside the Earth where shale gas is available
- (c) Search for the Earth-like planets in outer space
- (d) Search for meteorites containing precious metals

EXPLANATION

While the options a and c both seem plausible, the correct answer is the most appropriate answer. Option a is the definition of Goldilocks zone whereas in the context of it appearing in the news the correct answer should be Option c.

Exoplanet Missions



POSSIBLE MAINS QUESTION

TOPIC:

GS Paper III : Awareness in space; Latest developments in S&T

What is known by the term goldilocks zone? Explain the significance of the TRAPPIST-1 discovery in this context.

HINTS

1 Describe goldilocks zone.

2 Describe about TRAPPIST1 system- TRAPPIST, Spitzer telescope and Hubble Space Telescope

3 Significant “biosignatures” in exoplanets.

4 Viability of life in the three earth-like planets of TRAPPIST.

5 Understanding the origin of Earth with the help of exoplanets.(Evidence of Earth’s interiors in Geomorphology)

6 One small step for man becomes a huge leap for mankind

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